

BGlab1（part3+附加实验）

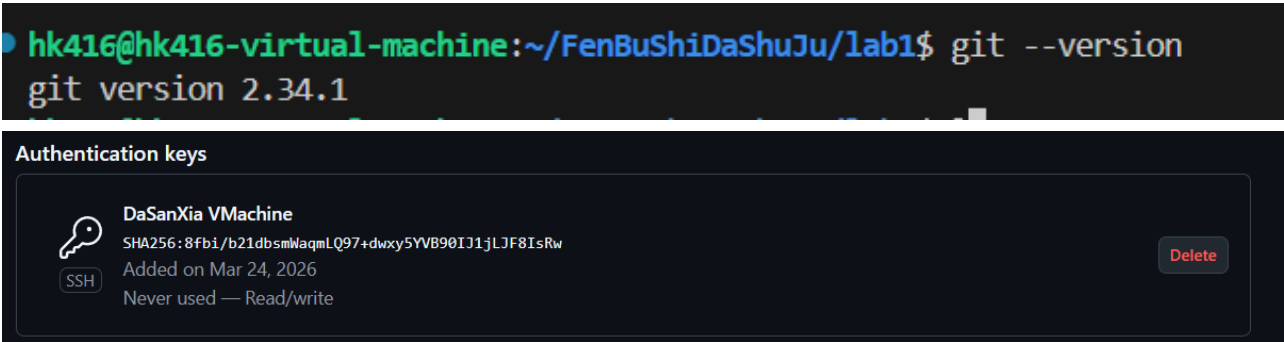
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学 号	231220069
日 期	2026 年 4 月 5 日

一、实验环境

- 操作系统：Ubuntu 22.04（WSL2）
- JDK 版本：OpenJDK 1.8.0_281
- Hadoop 版本：3.2.1
- 硬件配置：vCPU2核，内存4GB，硬盘30G

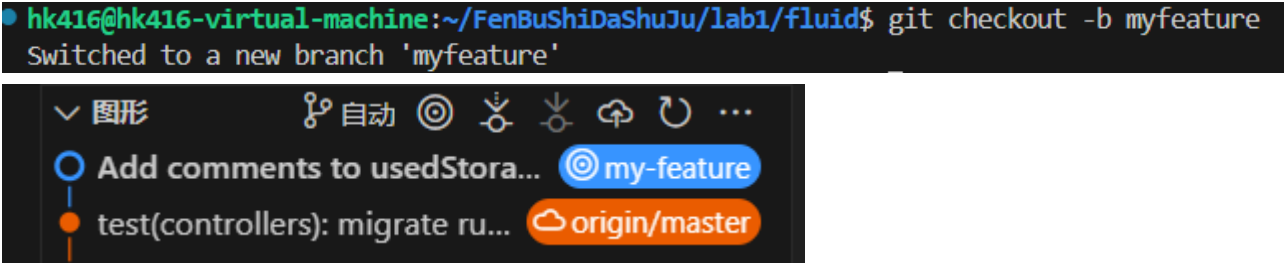
二、实验内容与步骤

1. Git 安装与配置，私钥的设置（完成截图如下）



2. Fork与Clone

- 在github上完成fork后克隆到本地(截图如下):



3. 代码修改与提交

a.为 pkg/ddc/alluxio/alluxio.go 中的 `usedStorageBytesInternal` 添加注释:

```
// usedStorageBytesInternal returns the number of bytes currently used by Alluxio storage.
```

```
// This method is intended for internal use by the AlluxioEngine.
// It currently returns (0, nil) as a placeholder; the actual implementation
// should query
// the Alluxio cluster to compute the used storage capacity.
func (e *AlluxioEngine) usedStorageBytesInternal() (value int64, err error) {
    return
}
```

```
// usedStorageBytesInternal returns the number of bytes currently used by Alluxio storage.
// This method is intended for internal use by the AlluxioEngine.
// It currently returns (0, nil) as a placeholder; the actual implementation should query
// the Alluxio cluster to compute the used storage capacity.
func (e *AlluxioEngine) usedStorageBytesInternal() (value int64, err error) {
    return
}
```

b. 提交修改

```
● hk416@hk416-virtual-machine:~/FenBuShiDaShuJu/lab1/fluid$ git push origin my-feature
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 754 bytes | 754.00 KiB/s, done.
Total 6 (delta 5), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (5/5), completed with 5 local objects.
remote:
remote: Create a pull request for 'my-feature' on GitHub by visiting:
remote:   https://github.com/HK416416/fluid/pull/new/my-feature
remote:
To github.com:HK416416/fluid.git
 * [new branch]      my-feature -> my-feature
```

4. 创建 Issue

- 在源仓库 fluid-cloudnative/fluid 创建 Issue, 描述本次添加注释的请求
- issue编号: 5716

← Create new issue in fluid-cloudnative/fluid

Add a title *

Add comments to usedStorageBytesInternal in fluid/pkg/ddc/alluxio/ufs_internal.go

Add a description

WritePreview

H B I  < >     @   ↩

Which function do you want to add comments to?

usedStorageBytesInternal in fluid/pkg/ddc/alluxio/ufs_internal.go

What comments do you want to add?

// usedStorageBytesInternal returns the number of bytes currently used by Alluxio storage.
// This method is intended for internal use by the AlluxioEngine.
// It currently returns (0, nil) as a placeholder; the actual implementation should query
// the Alluxio cluster to compute the used storage capacity.

 Paste, drop, or click to add files

☐ Create more

Cancel

Create 

 HK416416 opened now

Which function do you want to add comments to?

usedStorageBytesInternal in fluid/pkg/ddc/alluxio/ufs_internal.go

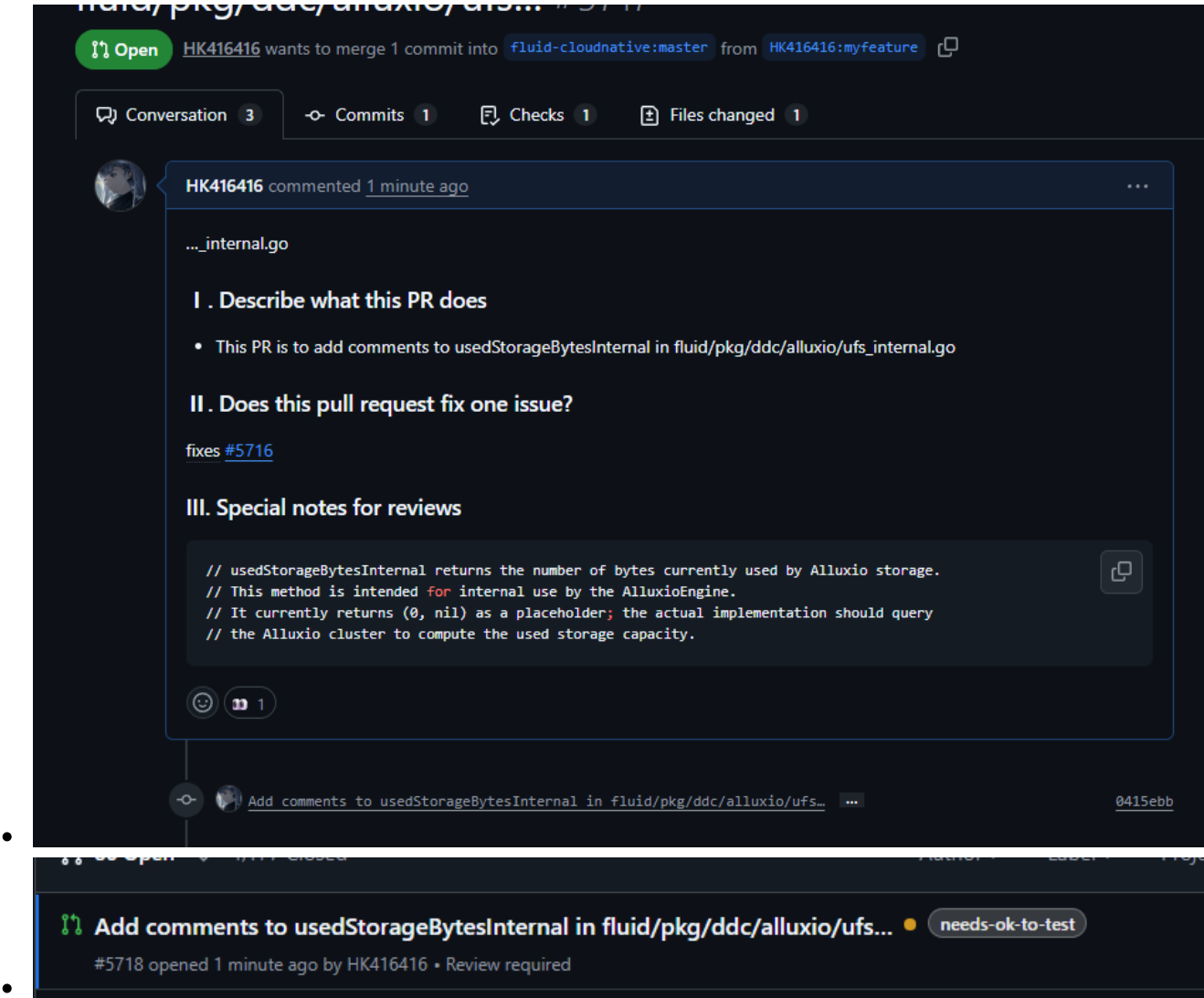
What comments do you want to add?

// usedStorageBytesInternal returns the number of bytes currently used by Alluxio storage.
// This method is intended for internal use by the AlluxioEngine.
// It currently returns (0, nil) as a placeholder; the actual implementation should query
// the Alluxio cluster to compute the used storage capacity.



5. 创建 Pull Request (PR)

- 创建PR并关联对应的issue
- PR编号: [5718](#)



6.issue关闭，PR合并

- 在PR被接受之后，issue也关闭

Add comments to usedStorageBytesInternal in fluid/pkg/ddc/alluxio/ufs_internal.go #5716

Closed

#5718 HK416416/fluid#1

HK416416 opened yesterday

Contributor

...

Which function do you want to add comments to?

usedStorageBytesInternal in fluid/pkg/ddc/alluxio/ufs_internal.go

What comments do you want to add?

// usedStorageBytesInternal returns the number of bytes currently used by Alluxio storage.
// This method is intended for internal use by the AlluxioEngine.
// It currently returns (0, nil) as a placeholder; the actual implementation should query
// the Alluxio cluster to compute the used storage capacity.

HK416416 mentioned this in 3 pull requests yesterday

Add comments to usedStorageBytesInternal in fluid/pkg/ddc/alluxio/ufs... HK416416/fluid#1

Add comments to usedStorageBytesInternal in fluid/pkg/ddc/alluxio/ufs... #5717

Add comments to usedStorageBytesInternal in fluid/pkg/ddc/alluxio/ufs... #5718

fluid-e2e-bot closed this as completed in #5718 11 hours ago

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Code with agent

- PR合并

Add comments to usedStorageBytesInternal in fluid/pkg/ddc/alluxio/ufs... #5718

Code

Merged

fluid-e2e-bot[bot] merged 1 commit into fluid-cloudnative:master from HK416416:my-feature 11 hours ago

Conversation 6

Commits 1

Checks 3

Files changed 1

+5

HK416416 commented yesterday

Contributor

...

..._internal.go

I . Describe what this PR does

Add comments to usedStorageBytesInternal in fluid/pkg/ddc/alluxio/ufs_internal.go

Signed-off-by: HK416416 1377784915@qq.com

II . Does this pull request fix one issue?

fixes #5716

III. Special notes for reviews

// usedStorageBytesInternal returns the number of bytes currently used by Alluxio storage.
// This method is intended for internal use by the AlluxioEngine.
// It currently returns (0, nil) as a placeholder; the actual implementation should query
// the Alluxio cluster to compute the used storage capacity.

Reviewers

cheyang

+1 more reviewer

gemini-code-assist[bot]

Assignees

cheyang

Labels

approved

lgtn

needs-ok-to-test

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues

三、附加实验

1.本地PI程序运行时间

- 时间： 34.113s

5 / 6

```
hk416@hk416-virtual-machine:~/hadoop_installs/hadoop-3.2.1$ hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.2.1.jar pi 8 1000000
Peak Map Virtual memory (bytes)=2524508160
Peak Reduce Physical memory (bytes)=178335744
Peak Reduce Virtual memory (bytes)=2531487744
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=944
File Output Format Counters
Bytes Written=97
Job Finished in 34.113 seconds
2026-03-24 15:36:40,739 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
Estimated value of Pi is 3.14159800000000000000
```

2.集群运行时间

- 时间：16.789s

```
Peak Map Virtual memory (bytes)=2681860096
Peak Reduce Physical memory (bytes)=318300160
Peak Reduce Virtual memory (bytes)=2686308352
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=944
File Output Format Counters
Bytes Written=97
Job Finished in 16.798 seconds
2026-04-07 02:25:50,443 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localhostTrusted = false, remoteHostTrusted = false
Estimated value of Pi is 3.14159800000000000000
```

3.分析区别原因

- 多个 Map 任务同时运行在不同节点上，充分利用多台机器的 CPU、内存和 I/O 资源，而本地伪分布式模式受限于单机资源，任务实际上是串行执行的。
- 但是加速比没有达到理想效果，可能是分布式处理增加了一些数据传输的耗时等。

4.本地grep运行以及结果

- example.txt文件内容如下

```
hadoop is a big data framework
I love hadoop
hadoop hadoop hadoop
```

```
hk416@hk416-virtual-machine:~/FenBuShiDaShuJu/lab1$ hdfs dfs -cat /user/hk416/grep-out/part-r-00000
hadoop
```